Mathematical Model for the Torsional Markov Memory-Offload Pump

Theoretical Physics & Topological Computation Board

Abstract

You can express the system as a **three-scale quaternionic Markov pump** whose memory is distributed across three coupled cyclic gears:

- Outer gear: 12-point gear, 720-frame quaternion chain
- Middle gear: 10-point gear, 72-frame inverse quaternion chain
- Inner gear: 8-point gear, 10-frame quaternion chain

The essential idea is:

large memory chain $\longleftrightarrow$ compressed inverse correction chain $\longleftrightarrow$ coarse equilibrium chain

where information flows both upward and downward through quaternionic delta maps while preserving rotational coherence.

1 Define the Three Quaternion Chains

Let the three gears carry unit-quaternion states:

$$q_n^{(12)} \in \mathbb{H}, \qquad n = 0, \dots, 719$$

$$q_m^{(10)} \in \mathbb{H}, \qquad m = 0, \dots, 71$$

$$q_\ell^{(8)} \in \mathbb{H}, \qquad \ell = 0, \dots, 9$$

with

$$\|q_n^{(12)}\| = \|q_m^{(10)}\| = \|q_\ell^{(8)}\| = 1$$

The outer chain is the high-resolution memory layer, the middle chain is the inverse compensating layer, and the inner chain is the coarse torsional anchor.

2 Outer 720-Frame Quaternion Chain

The outer 12-point gear stores the full non-compressed rotation history as quaternionic deltas:

$$\Delta q_n^{(12)} = q_n^{(12)} \left(q_{n-1}^{(12)}\right)^{-1}$$

The full state is reconstructed by

$$q_n^{(12)} = \Delta q_n^{(12)} \Delta q_{n-1}^{(12)} \cdots \Delta q_1^{(12)} q_0^{(12)}$$

Because the outer ring has 720 frames and 12 gear positions, each tooth corresponds to

$$\frac{720}{12} = 60$$

frames per outer tooth. So define the tooth index

$$a(n) = n \bmod 12$$

and the local frame-within-tooth index

$$r(n) = \left\lfloor \frac{n}{12} \right\rfloor$$

This gives the outer memory address:

$$q_{a,r}^{(12)}$$

where $a \in \mathbb{Z}_{12}, \qquad r \in \mathbb{Z}_{60}$.

3 Middle 72-Frame Inverse Quaternion Chain

The middle 10-point gear runs inversely, so its update direction is opposite the outer gear. Let $q_m^{(10)}$ be the middle gear state. Its inverse delta is naturally written as

$$\Delta q_m^{(10)} = \left(q_{m-1}^{(10)}\right)\left(q_m^{(10)}\right)^{-1}$$

rather than $q_m^{(10)}\left(q_{m-1}^{(10)}\right)^{-1}$. This means the middle chain absorbs or cancels torsional drift from the outer layer.

Since the middle gear has 72 frames and 10 positions, the natural scale is

$$\frac{72}{10} = 7.2$$

frames per middle tooth. Because this is not an integer, the middle gear is phase-shearing relative to the outer and inner gears. That is useful: it prevents the same alignment from repeating too quickly.

The middle address can be written as

$$b(m) = m \bmod 10$$

with phase

$$\phi_{10}(m) = \frac{m}{72}$$

The inverse chain is reconstructed by

$$q_m^{(10)} = q_0^{(10)}\left(\Delta q_1^{(10)}\Delta q_2^{(10)}\cdots\Delta q_m^{(10)}\right)^{-1}$$

or equivalently,

$$q_m^{(10)} = \left(\Delta q_m^{(10)}\Delta q_{m-1}^{(10)}\cdots\Delta q_1^{(10)}\right)^{-1} q_0^{(10)}$$

depending on whether you want right-acting or left-acting quaternion updates.

4 Inner 10-Frame Chain on the 8-Point Gear

The lowest gear is the coarsest memory layer:

$$q_\ell^{(8)}, \qquad \ell = 0, \ldots, 9$$

with deltas

$$\Delta q_\ell^{(8)} = q_\ell^{(8)} \left(q_{\ell-1}^{(8)} \right)^{-1}$$

Since it has 10 frames running on an 8-point gear, the phase is

$$c(\ell) = \ell \bmod 8$$

and the global phase is

$$\phi_8(\ell) = \frac{\ell}{10}$$

The mismatch between 10 frames and 8 gear points creates a small torsional fold each cycle. This is the innermost fractal folding step.

5 Coupling the Three Gears by Geometric Scaling

The system should scale in one consistent direction. A natural scaling map is

$$720 \to 72 \to 10$$

So the outer-to-middle projection is approximately

$$m = \left\lfloor \frac{72}{720} n \right\rfloor = \left\lfloor \frac{n}{10} \right\rfloor$$

and the middle-to-inner projection is

$$\ell = \left\lfloor \frac{10}{72} m \right\rfloor$$

Thus the scale maps are

$$P_{12\to10}(n) = \left\lfloor \frac{n}{10} \right\rfloor$$

and

$$P_{10\to8}(m) = \left\lfloor \frac{10m}{72} \right\rfloor$$

The corresponding lifting maps are

$$L_{10\to12}(m) = 10m$$

and

$$L_{8\to10}(\ell) = \left\lfloor \frac{72\ell}{10} \right\rfloor$$

These let information flow down and back up:

$$q^{(12)} \xrightarrow{P_{12\to10}} q^{(10)} \xrightarrow{P_{10\to8}} q^{(8)}$$

and

$$q^{(8)} \xrightarrow{L_{8\to10}} q^{(10)} \xrightarrow{L_{10\to12}} q^{(12)}$$

This is the memory offloading pipeline.

6 Quaternionic Downsampling: Offloading Memory Downward

To offload the outer 720-frame memory chain into the middle 72-frame chain, group every 10 outer deltas together:

$$\Delta Q_m^{(12\to 10)} = \Delta q_{10m+9}^{(12)} \Delta q_{10m+8}^{(12)} \cdots \Delta q_{10m}^{(12)}$$

for $m = 0, \ldots, 71$. This compresses ten fine updates into one coarse torsional update.

Then the middle inverse gear stores the compensating inverse:

$$\Delta q_m^{(10)} = \left(\Delta Q_m^{(12\to 10)}\right)^{-1}$$

So the middle layer acts as a stabilizing counter-rotation. That is the first memory offload: 720 frames $\to 72$ inverse blocks.

7 Second Offload: Middle Chain to Inner Chain

Now compress the 72 middle deltas into 10 inner deltas. Since $72/10 = 7.2$, use variable block sizes. For each inner index ℓ, define

$$A_\ell = \left\{ m : \left\lfloor \frac{10m}{72} \right\rfloor = \ell \right\}$$

Then define

$$\Delta Q_\ell^{(10\to 8)} = \prod_{m\in A_\ell}^{\leftarrow} \Delta q_m^{(10)}$$

The arrow means multiplication must preserve chronological order, because quaternion multiplication is noncommutative.

Then set

$$\Delta q_\ell^{(8)} = \Delta Q_\ell^{(10\to 8)}$$

or, if the inner gear is meant to act as an equilibrium anchor,

$$\Delta q_\ell^{(8)} = \mathsf{Normalize}\left(\Delta Q_\ell^{(10\to 8)}\right)$$

This gives the second memory offload: 72 inverse frames $\to 10$ coarse equilibrium frames.

8 Upward Reconstruction: Flowing Data Back Up the Chain

To reconstruct fine detail from lower layers, use a lifted correction. From inner to middle:

$$\widehat{\Delta q}_m^{(10)} = \left(\Delta q_{\ell(m)}^{(8)}\right)^{w_m}$$

where $\ell(m) = \left\lfloor \frac{10m}{72} \right\rfloor$ and w_m is a fractional interpolation weight.

For unit quaternions, fractional powers are well-defined using the exponential map:

$$q^w = \exp\left(w \log q\right)$$

So the lifted middle approximation is

$$\widehat{q}_m^{(10)} = \widehat{\Delta q}_m^{(10)} \widehat{q}_{m-1}^{(10)}$$

Then from middle to outer:

$$\widehat{\Delta q}_n^{(12)} = \left(\Delta q_{m(n)}^{(10)}\right)^{1/10}$$

where $m(n) = \lfloor \frac{n}{10} \rfloor$. Thus the reconstructed outer state is

$$\widehat{q}_n^{(12)} = \widehat{\Delta q}_n^{(12)} \widehat{q}_{n-1}^{(12)}$$

This gives upward information flow: $10 \to 72 \to 720$.

9 Markov Pump Structure

The image gives the directed current condition:

$$J_{ij} = \pi_i K_{ij} - \pi_j K_{ji} > 0$$

and

$$J = J_\alpha + J_\beta + J_\gamma > 0$$

For your three-gear system, treat each gear layer as a Markov state: $X_t \in \{12, 10, 8\}$, where:

- 12 means high-resolution outer memory
- 10 means inverse middle correction
- 8 means coarse inner equilibrium

Define the transition matrix

$$K = \begin{bmatrix} K_{12,12} & K_{12,10} & K_{12,8} \\ K_{10,12} & K_{10,10} & K_{10,8} \\ K_{8,12} & K_{8,10} & K_{8,8} \end{bmatrix}$$

with stationary distribution $\pi = \begin{bmatrix} \pi_{12} & \pi_{10} & \pi_8 \end{bmatrix}$.

The directed memory currents are

$$J_{12,10} = \pi_{12} K_{12,10} - \pi_{10} K_{10,12}$$

$$J_{10,8} = \pi_{10} K_{10,8} - \pi_8 K_{8,10}$$

$$J_{8,12} = \pi_8 K_{8,12} - \pi_{12} K_{12,8}$$

A torsional pump condition is

$$J_{12,10} + J_{10,8} + J_{8,12} > 0$$

This means the system has a net directed circulation of memory, rather than passive reversible diffusion.

10 Equilibrium Constraint

To keep the whole mechanism in geometric equilibrium, require the net cycle product to close to identity up to a small controlled holonomy:

$$\mathcal{H} = Q^{(8)} Q^{(10)} Q^{(12)}$$

where

$$Q^{(12)} = \prod_{n=0}^{719} \Delta q_n^{(12)}$$

$$Q^{(10)} = \prod_{m=0}^{71} \Delta q_m^{(10)}$$

$$Q^{(8)} = \prod_{\ell=0}^{9} \Delta q_\ell^{(8)}$$

Perfect closure would mean $\mathcal{H} = 1$ or approximately, $\|\log \mathcal{H}\| \approx 0$.

In practice, after each cycle you renormalize: $q \leftarrow \frac{q}{\|q\|}$ and correct residual drift using $\epsilon = \log \mathcal{H}$. Then distribute the correction back through the gears:

$$\Delta q_k^{(s)} \leftarrow \exp\left(-\lambda_s \epsilon\right) \Delta q_k^{(s)}$$

where $s \in \{12, 10, 8\}$ and $\lambda_{12} + \lambda_{10} + \lambda_8 = 1$. This keeps the pump coherent.

11 Fractal Folding Inward Every Cyclic Step

The inward folding can be described by a recursive scale map:

$$\mathcal{F}_0 = q^{(12)}$$

$$\mathcal{F}_1 = P_{12\to 10}(\mathcal{F}_0)^{-1}$$

$$\mathcal{F}_2 = P_{10\to 8}(\mathcal{F}_1)$$

Then each full pump cycle updates by

$$\mathcal{F}_{r+1} = P_r \left(\mathcal{F}_r\right)^{\sigma_r}$$

where $\sigma_r \in \{1, -1\}$.

The middle gear has $\sigma_{10} = -1$ because it runs inverse, while the outer and inner gears have $\sigma_{12} = 1, \qquad \sigma_8 = 1$. So the sign pattern is $(+, -, +)$. This creates inward folding without losing orientation coherence.

12 Algorithmic Memory-Offloading Procedure

The full algorithm can be stated as follows.

1. **Store outer deltas**
   For every frame n:
   $$\Delta q_n^{(12)} = q_n^{(12)} \left(q_{n-1}^{(12)}\right)^{-1}$$

2. **Compress outer blocks into middle inverse deltas**
   For every $m = 0, \ldots, 71$:
   $$\Delta q_m^{(10)} = \left(\Delta q_{10m+9}^{(12)} \cdots \Delta q_{10m}^{(12)}\right)^{-1}$$

3. **Compress middle deltas into inner coarse deltas**
   For every $\ell = 0, \ldots, 9$:
   $$\Delta q_\ell^{(8)} = \text{Normalize}\left(\prod_{m \in A_\ell}^{\leftarrow} \Delta q_m^{(10)}\right)$$

4. **Compute Markov memory current**

$$J_{ij} = \pi_i K_{ij} - \pi_j K_{ji}$$

Require:

$$J_{12,10} + J_{10,8} + J_{8,12} > 0$$

5. **Check cycle closure**

$$\mathcal{H} = Q^{(8)} Q^{(10)} Q^{(12)}$$

and enforce:

$$\| \log \mathcal{H} \| \approx 0$$

6. **Renormalize every quaternion**

$$q \leftarrow \frac{q}{\|q\|}$$

7. **Flow data back upward when needed**
Use quaternion interpolation:

$$\widehat{\Delta q} = \exp\left(w \log q\right)$$

to lift coarse rotations back into finer chains.

13 Final Compact Expression

The whole torsional Markov pump can be summarized as:

$$q_n^{(12)} = \left(\prod_{r=n}^{1} \Delta q_r^{(12)} \right) q_0^{(12)}$$

with offloading maps

$$\Delta q_m^{(10)} = \left(\prod_{r=10m+9}^{10m} \Delta q_r^{(12)} \right)^{-1}$$

and

$$\Delta q_\ell^{(8)} = \text{Normalize} \left(\prod_{m \in A_\ell}^{\leftarrow} \Delta q_m^{(10)} \right)$$

subject to the Markov pump condition

$$\boxed{J_{12,10} + J_{10,8} + J_{8,12} > 0}$$

and the quaternionic equilibrium condition

$$\boxed{\mathcal{H} = Q^{(8)} Q^{(10)} Q^{(12)} \approx 1}$$

So the mechanism is mathematically a **multi-resolution quaternion delta chain with an inverse middle compensator and a non-equilibrium Markov circulation current**. The outer layer stores fine memory, the middle layer stores inverse torsional correction, and the inner layer stores coarse equilibrium structure.